

|                               |                                             |                                                             |                                                |
|-------------------------------|---------------------------------------------|-------------------------------------------------------------|------------------------------------------------|
| <b>Notice of Allowability</b> | <b>Application No.</b><br>19/394,177        | <b>Applicant(s)</b><br>SUBBARAMA, Raman<br>Bhagavatheeswara |                                                |
|                               | <b>Examiner</b><br>CARLOS R ORTIZ RODRIGUEZ | <b>Art Unit</b><br>2119                                     | <b>AIA (First Inventor to File) Status</b> Yes |

**-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--**

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 06/01/2026.  
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on \_\_\_\_\_.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on \_\_\_\_\_; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-17. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see [http://www.uspto.gov/patents/init\\_events/pph/index.jsp](http://www.uspto.gov/patents/init_events/pph/index.jsp) or send an inquiry to **PPHfeedback@uspto.gov**.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

**Certified copies:**

- a) ☐ All      b) ☐ Some\*      c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
  2. ☐ Certified copies of the priority documents have been received in Application No. \_\_\_\_\_.
  3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

\* Certified copies not received: \_\_\_\_\_.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

**THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.**

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.  
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date \_\_\_\_\_.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

**Attachment(s)**

- |                                                                                                                   |                                                                                      |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1. <input checked="" type="checkbox"/> Notice of References Cited (PTO-892)                                       | 5. <input type="checkbox"/> Examiner's Amendment/Comment                             |
| 2. <input type="checkbox"/> Information Disclosure Statements (PTO/SB/08),<br>Paper No./Mail Date _____.          | 6. <input checked="" type="checkbox"/> Examiner's Statement of Reasons for Allowance |
| 3. <input type="checkbox"/> Examiner's Comment Regarding Requirement for Deposit<br>of Biological Material _____. | 7. <input type="checkbox"/> Other _____.                                             |
| 4. <input type="checkbox"/> Interview Summary (PTO-413),<br>Paper No./Mail Date _____.                            |                                                                                      |

/CARLOS R ORTIZ RODRIGUEZ/  
Primary Examiner, Art Unit 2119

**DETAILED ACTION**

Claims 1-17 are pending.

***Notice of Pre-AIA or AIA Status***

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

***Allowable Subject Matter***

Claims 1-17 are allowed.

The following is a statement of reasons for the indication of allowable subject matter:

The prior art of record taken either alone or in combination fails to disclose a system/method comprising the following limitation/feature:

(Claim 1) - "(ii) evaluate each candidate state update using a stability potential, wherein a scalar function of system state representing deviation from the invariant relationships, wherein the CE selects a state update from the set of candidate state updates by minimizing the stability potential subject to invariance constraints derived from the invariant relationships detected by the Invariance Monitor (IM); and a Feedback Controller (FC) configured to apply the selected state update to the system and store results for subsequent evaluation, wherein the SD, IM, CE, and FC operate cooperatively in a recursive control cycle that: continuously evaluates invariant relationships, applies bounded state updates, and maintains coherence of system state over successive iterations without external reset, and wherein the

Art Unit: 2119

system operates independently of an underlying computational substrate through on normalized state descriptors", and

(Claim 2) – “evaluating each candidate state update using a stability potential, the stability potential being a scalar function representing deviation from the invariant relationships; (e) selecting a state update from the candidate state updates by minimizing the stability potential subject to invariance constraints derived from the detected invariant relationships; and (f) applying the selected state update to the system state in a bounded recursive feedback loop, wherein the method maintains operational identity of the system over successive iterations without external reset”,

in combination with the remaining elements and features of the claimed invention.

It is for these reasons that the applicant's invention defines over the prior art of record.